

Modeling High-Dimensional Outliers

- ❑ Develop new models for high-dimensional outliers directly, typically avoid proximity measures and instead adopt new heuristics to detect outliers, which do not deteriorate in high-dimensional data
- ❑ Example: angle-based outlier detection (ABOD)
 - ❑ Given a set of points, D , for a point, $o \in D$, define the angle-based outlier factor (ABOF) as $ABOF(o) = var_{x,y \in D \setminus \{o\}} \left(\frac{\langle \overrightarrow{ox}, \overrightarrow{oy} \rangle}{dist(o,x)^2 dist(o,y)^2} \right)$, where $\langle \overrightarrow{ox}, \overrightarrow{oy} \rangle$ is the scalar (dot) product
 - ❑ The farther away a point is from clusters and the smaller the variance of the angles of a point, the smaller the ABOF
 - ❑ Compute the ABOF for each point, and output a list of the points in the data set in ABOF-ascending order

ABOD Example

